Signed low-entropy reweightings to rank one on the sphere

A campaign report: Question 8.1 of Barak–Kothari–Steurer, answered

Conjura campaign `c/0048` · artifact `0048-RKHS-1-r5`

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Result. Question 8.1 of Barak–Kothari–Steurer (arXiv:1701.06321v2, p. 19) has an affirmative answer, and a much stronger one than it asks for: entropy cost $\log\bigl(n(n+1)/2\bigr) = O(\log n)$, Frobenius error exactly 0, and a constant independent of ε. The paper asks whether cost $O(n^\delta)$ is achievable for arbitrarily small $\delta > 0$; $O(\log n)$ is below every n^δ.

Verification status. Not certified. Four rounds of five independent blind referee passes (20 passes, 3 models, 5 angles) have been run. No pass in any round has faulted a load-bearing step; every substantive defect found has been in the scope commentary, and the load-bearing text is byte-identical across all five revisions. Best round scored 4 clean of 5; the current revision `r5` is unverified. Details in §7.

Caveat that matters. The passes were run as in-session subagents, not as the isolated `--bare` processes the harness specifies, because this host has no model backend. One vendor, three models. See §8.

1 The question

The source paper's Question 8.1 reads, verbatim (p. 19):

> *Is it the case that for every distribution μ over $\mathbb{S}^{n-1}$ and every $\varepsilon, \delta > 0$ there is a (not necessarily positive) function $r : \mathbb{S}^{n-1} \to \mathbb{R}$ such that $\mathbb{E}_{v\sim\mu}|r(v)| = 1$, $\mathbb{E}_{v\sim\mu}|r(v)|\log|r(v)| \leqslant O(n^\delta)$ and a nonzero rank one L such that $\|\mathbb{E}_{v\sim\mu}[r(v)vv^\top] - L\|_F \leqslant \varepsilon\|L\|_F$?*

The paper adds: *"A positive solution for Question 8.1 for any $\delta < 1/2$ would be very interesting. It may improve the best known bound for the log rank conjecture to $\tilde{O}(n^\delta)$ and if appropriately extended to pseudo-distributions, improve our algorithm's running time to* $\exp(\tilde{O}(n^\delta))$ *as well. We do know that the answer to this question is No if one does not allow negative reweighting functions."*

The formal statement attacked here was transcribed independently from that PDF and then checked against it; the transcription is faithful, with no additional hypothesis. Call a Borel measurable, μ-integrable $r : \mathbb{S}^{n-1} \to \mathbb{R}$ — *not* required nonnegative — a *signed reweighting of μ with entropy cost at most K* if

$$\mathbb{E}_{v\sim\mu}|r(v)| = 1, \qquad \mathbb{E}_{v\sim\mu}\bigl[\,|r(v)|\log|r(v)|\,\bigr] \le K, \qquad \text{convention } 0\log 0 := 0.$$

Where the difficulty was expected to lie. Write $|r| = w$ and $\sigma = \operatorname{sign} r$. Then $\mathbb{E}_\mu|r| = 1$ makes $d\nu = w\,d\mu$ a probability measure with $\Delta_{\mathrm{KL}}(\nu\|\mu) = \mathbb{E}_\mu[w\log w]$, and $\mathbb{E}_\mu[r\,vv^\top] = \mathbb{E}_\nu[\sigma(v)\,vv^\top]$. So the question asks: tilt μ at entropy price K, then choose signs freely, and make the resulting second moment relatively close to rank one. In the nonnegative case $\sigma \equiv 1$ the matrix $\mathbb{E}_\nu[vv^\top]$ is PSD of trace 1, so $\|\mathbb{E}_\nu[vv^\top]\|_F \ge n^{-1/2}$: the target cannot be shrunk, and that floor is what makes the nonnegative problem cost $\sqrt{n}$. Signs remove the floor. The proof below removes it completely — it makes the second moment *exactly* a rank one matrix.

2 The result

Theorem 1. *For every $n \geq 1$ and every Borel probability measure μ on $\mathbb{S}^{n-1}$ there exist a point $v_0 \in \operatorname{supp}\mu$, a constant c with $\big(n(n+1)/2\big)^{-1/2} \leq c \leq 1$, and a signed reweighting r of μ — indeed the restriction to $\mathbb{S}^{n-1}$ of a single quadratic form, hence a bounded Borel function — such that*

$$\mathbb{E}_{v\sim\mu}\big[\, r(v)\, vv^\top \big] \;=\; c\, v_0 v_0^\top \qquad \text{exactly},$$

and whose entropy cost satisfies

$$\mathbb{E}_\mu\big[\, |r| \log |r| \,\big] \;\leq\; \log K(v_0, v_0) \;\leq\; \log\big(n(n+1)/2\big).$$

Here $L := c\, v_0 v_0^\top$ is rank one, nonzero and PSD, with $\|L\|_F = c$.

Corollary 2. *Question 8.1 holds with $C(\varepsilon, \delta) := 2/(e\delta)$, a constant independent of ε: for every $\varepsilon > 0$, $\delta > 0$, every n and every Borel probability μ on $\mathbb{S}^{n-1}$ there are a signed reweighting of entropy cost at most $\log(n(n+1)/2) \leq (2/(e\delta))\, n^\delta$ and a nonzero rank one L with $\|\mathbb{E}_\mu[r(v)vv^\top] - L\|_F = 0 \leq \varepsilon \|L\|_F$.*

The mechanism, in one sentence. On the sphere the constant function 1 *is* a quadratic form, namely $v \mapsto v^\top I v = \|v\|_2^2$; so the reproducing kernel of the quadratic forms inside $L^2(\mu)$ has mean exactly 1, which normalises it for free, and it reproduces every coordinate product $v_a v_b$, which makes its second moment exactly $v_0 v_0^\top$. The entropy price is then the logarithm of its squared L^2 norm, which averages over μ to the dimension of the space, at most $n(n+1)/2$.

3 The proof

Throughout, $\mathbb{S}^{n-1}$ carries its usual compact, second-countable metric topology; μ is a Borel probability measure on it; Sym_n denotes the real symmetric $n \times n$ matrices; and for $Q \in \mathrm{Sym}_n$ we write $q_Q(v) = v^\top Q v$. Two facts about the support are used and recorded now.

(P1) $\mu(\operatorname{supp}\mu) = 1$. The complement of the support is a union of open null sets; since $\mathbb{S}^{n-1}$ is second countable that union is a countable union of null sets, hence null.

(P2) If $v \in \operatorname{supp}\mu$ and $U \ni v$ is open, then $\mu(U) > 0$. This is the definition of the support.

Lemma 3 (the quadratic space, evaluation on the support, and its kernel)**.** *Let $F_\mu := \{\, [q_Q] \in L^2(\mu) : Q \in \mathrm{Sym}_n \,\}$, where $[\cdot]$ is the μ-a.e. class, and let $d := \dim F_\mu$. Then*

(a) F_μ is a linear subspace of $L^\infty(\mu) \cap L^2(\mu)$ with $d \leq n(n+1)/2$ and $\|q_Q\|_\infty \leq \|Q\|_F$;

(b) if $q_Q = q_{Q'}$ holds μ-a.e. then $q_Q(v) = q_{Q'}(v)$ for every *$v \in \operatorname{supp}\mu$; hence for $v \in \operatorname{supp}\mu$ the map $\mathrm{ev}_v : F_\mu \to \mathbb{R}$, $[q_Q] \mapsto q_Q(v)$, is well defined and linear;*

(c) for each $v \in \operatorname{supp}\mu$ there is a unique $k_v \in F_\mu$ with $\langle k_v, f\rangle_{L^2(\mu)} = \mathrm{ev}_v(f)$ for all $f \in F_\mu$; set $K(v,v) := \|k_v\|^2_{L^2(\mu)} = \mathrm{ev}_v(k_v)$;

(d) $[1] = [q_I] \in F_\mu$, and $[v \mapsto v_a v_b] = [q_{E_{ab}}] \in F_\mu$ where $E_{ab} = (e_a e_b^\top + e_b e_a^\top)/2$.

Proof. (a) $Q \mapsto q_Q$ is linear, so F_μ is the image of Sym_n and $d \leq \dim \mathrm{Sym}_n = n(n+1)/2$. For $v \in \mathbb{S}^{n-1}$, $|v^\top Q v| \leq \|Q\|_{\mathrm{op}} \|v\|_2^2 = \|Q\|_{\mathrm{op}} \leq \|Q\|_F$. Since μ is a probability measure, $L^\infty(\mu) \subseteq L^2(\mu)$.

(b) Put $h := q_{Q-Q'}$, a polynomial and hence continuous, with $\mu(\{h \neq 0\}) = 0$. Suppose $h(v_0) \neq 0$ for some $v_0 \in \operatorname{supp}\mu$. Then $U := \{|h| > |h(v_0)|/2\}$ is open and contains v_0, so $\mu(U) > 0$ by (P2); but $U \subseteq \{h \neq 0\}$ is null — a contradiction. Hence $h \equiv 0$ on $\operatorname{supp}\mu$. Well-definedness of ev_v follows: ev_v is defined through quadratic-form representatives only, and by the displayed statement any two quadratic-form representatives of the same class of F_μ take the same value at every $v \in \operatorname{supp}\mu$; linearity is inherited from $Q \mapsto q_Q(v)$.

(c) F_μ is a finite-dimensional subspace of $L^2(\mu)$, hence complete, hence a Hilbert space; ev_v is a linear functional on it, hence bounded, so Riesz representation supplies a unique $k_v \in F_\mu$. Explicitly $k_v = \sum_j \mathrm{ev}_v(f_j) f_j$ for any orthonormal basis (f_j). Taking $f = k_v$ gives $K(v,v) = \mathrm{ev}_v(k_v)$.

(d) $q_I(v) = \|v\|_2^2 = 1$ identically on $\mathbb{S}^{n-1}$, and $q_{E_{ab}}(v) = v_a v_b$ identically. □

Part (b) is the only place where the general Borel case differs from the finitely supported one, and it is what removes the need for any approximation argument: for finitely supported μ it is trivial, every support point being an atom.

Lemma 4 (three reproducing identities)**.** *Let $v_0 \in \operatorname{supp}\mu$ and $k := k_{v_0}$. Then*

(i) $\mathbb{E}_\mu[\, k(v)\, vv^\top \,] = v_0 v_0^\top$*;*

(ii) $\mathbb{E}_\mu[k] = 1$*; hence* $\mathbb{E}_\mu|k| \geq 1$*,* $K(v_0,v_0) \geq 1$*, and* $k \neq 0$*;*

(iii) $\mathbb{E}_\mu[k^2] = K(v_0,v_0)$ *and* $1 \leq \mathbb{E}_\mu|k| \leq K(v_0,v_0)^{1/2} < \infty$*.*

Proof. All integrals converge absolutely: k is bounded by Lemma 3(a), and the other factors are bounded by 1 on $\mathbb{S}^{n-1}$.

(i) Fix a, b. By Lemma 3(d) the class $g_{ab} := [v \mapsto v_a v_b]$ lies in F_μ, so the reproducing property gives $\mathbb{E}_\mu[k\, v_a v_b] = \langle k, g_{ab}\rangle = \mathrm{ev}_{v_0}(g_{ab}) = (v_0)_a (v_0)_b$. The (a,b) entries of the two matrices agree, for all a, b.

(ii) By Lemma 3(d), $[1] \in F_\mu$ with $\mathrm{ev}_{v_0}([1]) = q_I(v_0) = 1$, so $\mathbb{E}_\mu[k] = \langle k, [1]\rangle = 1$. Then $\mathbb{E}_\mu|k| \geq |\mathbb{E}_\mu k| = 1$ and, by Cauchy–Schwarz, $K(v_0,v_0) = \mathbb{E}_\mu[k^2] \geq (\mathbb{E}_\mu|k|)^2 \geq 1$. And $\mathbb{E}_\mu[k] = 1 \neq 0$ forces $k \neq 0$.

(iii) $K(v_0,v_0) = \|k\|^2 = \mathbb{E}_\mu[k^2]$ by definition; finiteness is Lemma 3(a); the upper bound is Cauchy–Schwarz against 1. □

Statement (ii) is precisely the trace of (i), since $\operatorname{tr}\mathbb{E}_\mu[k\, vv^\top] = \mathbb{E}_\mu[k\|v\|_2^2] = \mathbb{E}_\mu[k]$ while $\operatorname{tr}(v_0 v_0^\top) = 1$. This locates where the construction gets its normalisation for free: the sphere constraint $\|v\|_2 = 1$ is what makes the constant function a quadratic form. On a domain where $\|v\|_2$ were not constant, neither $[1] \in F_\mu$ nor $\mathbb{E}_\mu[k] = 1$ would be available.

Lemma 5 (a cheap base point exists)**.** *$\int K(v,v)\, d\mu(v) = d$, and consequently there exists $v_0 \in \operatorname{supp}\mu$ with $K(v_0,v_0) \leq d \leq n(n+1)/2$.*

Proof. Fix an orthonormal basis $f_1, \dots, f_d$ of F_μ and quadratic-form representatives q_j of f_j. For $v \in \operatorname{supp}\mu$, Lemma 3(c) and Parseval give $k_v = \sum_j q_j(v) f_j$, hence

$$K(v,v) \;=\; \sum_{j=1}^{d} q_j(v)^2 \;=:\; g(v) \qquad \text{for all } v \in \operatorname{supp}\mu. \tag{1}$$

Now g is a polynomial, hence Borel, and by (P1) it represents $v \mapsto K(v,v)$ up to a null set, so the integral is well defined and

$$\int K(v,v)\, d\mu \;=\; \sum_j \int q_j^2\, d\mu \;=\; \sum_j \|f_j\|^2 \;=\; d. \tag{2}$$

(For $v \in \operatorname{supp}\mu$ the value $g(v) = \|k_v\|^2$ is independent of the choice of basis and of representatives, by uniqueness in Lemma 3(c); off the support nothing below uses g.)

Let $A := \{g \leq d\}$, a Borel set. If $\mu(A) = 0$ then $h := g - d$ satisfies $h > 0$ μ-a.e. while $\int h\, d\mu = 0$ by (2); a nonnegative function with vanishing integral vanishes a.e., so $h = 0$ μ-a.e., which cannot hold simultaneously with $h > 0$ μ-a.e. under a probability measure. So $\mu(A) > 0$, hence by (P1) $A \cap \operatorname{supp}\mu \neq \emptyset$, and any v_0 in it satisfies $K(v_0,v_0) = g(v_0) \leq d$. □

Identity (2) is the statement $\int K(v,v)\,d\mu = \operatorname{tr} P$ for P the orthogonal projection of $L^2(\mu)$ onto F_μ — equivalently, that the reciprocal Christoffel function has μ-mean d. It is proved above from scratch and depends on no external source.

Lemma 6 (entropy at most the log of the second moment)**.** *If $r \in L^2(\mu)$ and $\mathbb{E}_\mu|r| = 1$, then $\mathbb{E}_\mu[\,|r|\log|r|\,] \le \log \mathbb{E}_\mu[r^2]$.*

Proof. Write $\varphi(t) = t\log t$ for $t > 0$ and $\varphi(0) = 0$. Then $\varphi(t) \ge -1/e$ for $t \ge 0$, and $\varphi(t) \le t^2$ for all $t \ge 0$ (for $t \ge 1$ because $\log t \le t$; for $t < 1$ because $\varphi \le 0$). Since $r \in L^2(\mu)$ and μ is a probability measure, $\varphi(|r|) \in L^1(\mu)$, so the entropy is a finite real number.

Let $d\nu := |r|\,d\mu$. Then $\nu(\mathbb{S}^{n-1}) = \mathbb{E}_\mu|r| = 1$, so ν is a probability measure, and $\nu(\{r = 0\}) = 0$, so $\log|r|$ is ν-a.e. finite and $\mathbb{E}_\nu[\log|r|] = \mathbb{E}_\mu[|r|\log|r|]$, the convention $0\log 0 = 0$ matching the ν-null set $\{r = 0\}$. Also $\mathbb{E}_\nu[|r|] = \mathbb{E}_\mu[r^2] < \infty$. Jensen's inequality for the concave function log against ν gives $\mathbb{E}_\nu[\log|r|] \le \log \mathbb{E}_\nu[|r|] = \log \mathbb{E}_\mu[r^2]$. □

Equality holds exactly when $|r|$ is ν-a.s. constant, i.e. when $|r| \in \{0, c\}$ μ-a.e., in which case the bound reads $\log(1/\mu(\{|r| = c\}))$. So Lemma 6 is tight precisely on the indicator-type reweightings — those for which "reweighting" means conditioning on an event — and is lossy exactly to the extent that $|r|$ is spread out.

Proof of Theorem 1

Let n and μ be given. Lemma 5 supplies $v_0 \in \operatorname{supp}\mu$ with $K(v_0,v_0) \le d \le n(n+1)/2$; put $k := k_{v_0}$. By Lemma 4(ii),(iii) we have $1 \le \mathbb{E}_\mu|k| \le K(v_0,v_0)^{1/2} < \infty$ and $k \ne 0$, so

$$c \;:=\; \frac{1}{\mathbb{E}_\mu|k|} \;\in\; \big[\,K(v_0,v_0)^{-1/2},\,1\big] \;\subseteq\; \big[\,(n(n+1)/2)^{-1/2},\,1\,\big]$$

is well defined and strictly positive. Put $r := c\,k$, with representative the quadratic form q_{cQ_k}; then r is a bounded Borel function on $\mathbb{S}^{n-1}$, hence μ-integrable and in $L^2(\mu)$, and $\mathbb{E}_\mu|r| = c\,\mathbb{E}_\mu|k| = 1$. So r is an admissible signed reweighting.

Second moment. By Lemma 4(i), $\mathbb{E}_\mu[r(v)vv^\top] = c\,v_0v_0^\top =: L$. Since $c > 0$ and $\|v_0\|_2 = 1$, $L = (cv_0)v_0^\top$ is rank one, nonzero and PSD with $\|L\|_F = c > 0$. The Frobenius error is $0 \le \varepsilon\|L\|_F$ for every $\varepsilon > 0$.

Entropy. Lemma 6 applies, and with Lemma 4(iii),

$$\begin{aligned}\mathbb{E}_\mu[|r|\log|r|] \;&\le\; \log\mathbb{E}_\mu[r^2] \;=\; \log\big(c^2K(v_0,v_0)\big) \;=\; \log\frac{K(v_0,v_0)}{(\mathbb{E}_\mu|k|)^2}\\ &\le\; \log K(v_0,v_0) \;\le\; \log\frac{n(n+1)}{2},\end{aligned}$$

using $\mathbb{E}_\mu|k| \ge 1$ from Lemma 4(ii) and Lemma 5. □

Proof of Corollary 2

For $n = 1$, $\log(n(n+1)/2) = \log 1 = 0$. For $n \ge 2$, $n(n+1)/2 \le n^2$ gives $\log(n(n+1)/2) \le 2\log n$; and $\max_{x>0}(\log x)/x^\delta = 1/(e\delta)$, attained at $x = e^{1/\delta}$, gives $2\log n \le (2/(e\delta))n^\delta$. Combine with Theorem 1. □

4 Two supporting statements

Remark 7 (the conclusion is not obtainable by cancelling to zero). Let $M \in \mathrm{Sym}_n$ have eigenvalues indexed so that $|\lambda_1| \ge |\lambda_2| \ge \cdots$, and let $\varepsilon \in (0,1)$. If $M = 0$, no admissible L exists, since $\|0 - L\|_F = \|L\|_F > \varepsilon\|L\|_F$ for every $L \ne 0$. If $M \ne 0$, a nonzero rank one L with $\|M - L\|_F \le \varepsilon\|L\|_F$ exists *if and only if* $\sum_{i\ge 2}\lambda_i^2 \le \big(\varepsilon^2/(1-\varepsilon^2)\big)\lambda_1^2$.

Proof. Write $L = t\,uw^\top$ with $\|u\| = \|w\| = 1$ and $t > 0$, so $\|L\|_F = t$ and $\|M - L\|_F^2 = \|M\|_F^2 - 2t\,u^\top Mw + t^2$. The requirement is $(1-\varepsilon^2)t^2 - 2(u^\top Mw)t + \|M\|_F^2 \le 0$, and increasing $u^\top Mw$ only helps, so the optimal choice is $u^\top Mw = s_1(M) = |\lambda_1|$ by the SVD. The quadratic in t has a real root iff $s_1^2 \ge (1-\varepsilon^2)\|M\|_F^2 = (1-\varepsilon^2)\sum_i \lambda_i^2$, which rearranges to the stated inequality; and when $M \ne 0$ both roots are positive, their product being $\|M\|_F^2/(1-\varepsilon^2) > 0$ and their sum $2s_1/(1-\varepsilon^2) > 0$, so an admissible $t > 0$ exists exactly then. □

This is not load-bearing. It is recorded because it shows that the relative normalisation $\|\cdot\|_F \le \varepsilon\|L\|_F$ does real work: the conclusion cannot be reached by cancelling the second moment to zero, so Theorem 1 is not exploiting a degenerate reading of the question.

Corollary 8 (the same for rank one matrices of unit Frobenius norm)**.** *Let $\mathcal{X} := \{X \in \mathbb{R}^{n\times n} : \operatorname{rank} X = 1,\ \|X\|_F = 1\}$, equivalently $X = ab^\top$ with $\|a\|_2 = \|b\|_2 = 1$ — the normalisation used in the source paper's intended application. For every Borel probability measure μ on $\mathcal{X}$ there exist $X_0 \in \operatorname{supp}\mu$, $c \in (0,1]$ and a signed reweighting r of μ with $\mathbb{E}_\mu[|r|\log|r|] \le \log(n^2+1)$ such that $\mathbb{E}_{X\sim\mu}[r(X)\,X] = c\,X_0$, exactly rank one and nonzero.*

Proof. Run Lemmas 3–6 verbatim with $\mathbb{S}^{n-1}$ replaced by the compact set $\mathcal{X}$ and the quadratic forms replaced by the affine space $G := \{X \mapsto \alpha + \langle A, X\rangle : \alpha \in \mathbb{R},\ A \in \mathbb{R}^{n\times n}\}$, of dimension at most n^2+1. Only two properties of the function space were ever used: it contains the constant 1 (here trivially, $\alpha = 1$, $A = 0$) and it contains each coordinate function $X \mapsto X_{ab}$ (here $\alpha = 0$, $A = e_a e_b^\top$). Continuity of the elements of G gives Lemma 3(b) unchanged, and boundedness on $\mathcal{X}$ gives Lemma 3(a). □

So the mechanism is not an artefact of specialising to $X = vv^\top$: what it needs is the presence of constants and coordinates in one finite-dimensional function space. On the sphere the homogeneous quadratic forms suffice because $\|v\|_2 = 1$ makes the constant a quadratic form, and that buys only the sharper bound $\log(n(n+1)/2)$ over $\log(n^2+1)$.

5 Scope: what this does and does not establish

Theorem 1 answers Question 8.1 as literally posed, and nothing more. Three delimitations, each checked against the source text.

(i) No conflict with the source's negative results. The remark that *"as stated, Theorem 2.3 is tight"* concerns *deficient* reweightings, which by the paper's Definition 2.2 are probability distributions and hence nonnegative; and *"the answer to this question is No if one does not allow negative reweighting functions"* is an existential statement over μ. Neither is contradicted here. Whenever the construction returns a nonnegative r, that r is itself a nonnegative reweighting of cost at most $\log(n(n+1)/2) = O(\log n)$ — more precisely, $k \ge 0$ forces $\mu(\{\pm v_0\}) = 1/K(v_0, v_0) \ge 2/(n(n+1))$ — so such a μ is not a hard instance for the nonnegative question. The same holds on the domain of Corollary 8, where the corresponding bound is the single-atom $\mu(\{X_0\}) = 1/K(X_0, X_0) \ge 1/(n^2+1)$. What is *not* addressed is the unnormalised domain of the paper's Theorem 2.3, "any distribution over rank one $n \times n$ matrices"; Corollary 8 treats only the unit-Frobenius slice, and nothing is claimed beyond it.

(ii) No log rank consequence is claimed. The paper's dictionary from reweightings to submatrices is stated only for *flat* reweightings — *"a flat distribution μ' with $\Delta_{\mathrm{KL}}(\mu'\|\mu) \leqslant k$ corresponds to the uniform distribution over $\{u_i v_i^\top\}_{i\in I}$ where $I \subseteq [N]$ satisfies $|I| \geqslant 2^{-k}N$"* — and Theorem 1 supplies no flatness guarantee, so that route is unavailable. No other route is addressed, and the paper's dual Theorem 2.4 is untouched. Morally the target $c\,v_0v_0^\top$ is the $|I| = 1$ answer; the content of Theorem 1 is that cancellation reaches *that same matrix* at cost $\log(n(n+1)/2)$, where conditioning would have cost $\log(1/\mu(\{v_0\}))$ — infinite for continuous μ.

(iii) The pseudo-distribution version is out of scope, and the obstruction is identifiable. The inner product used, $\langle q_Q, q_{Q'} \rangle = \mathbb{E}_\mu[(v^\top Qv)(v^\top Q'v)]$, is a degree-4 moment and does have a pseudo-distribution analogue. What has none is ev_{v_0}: a pseudo-distribution has no support points, and Lemma 3(b) — the step that makes point evaluation a well-defined functional on F_μ — is exactly what fails. Every use of $\operatorname{supp}\mu$ is therefore load-bearing for the honest-distribution case and unavailable in the setting the paper needs for its running-time consequence. Footnote 8's worry about the ε-dependence is moot here in the strongest way: there is none.

6 Worked example

An independent check of the three identities on a case where the kernel is not an indicator. Take $n = 2$ and μ uniform on $v_1 = (1, 0)$, $v_2 = (0, 1)$, $v_3 = (1, 1)/\sqrt{2}$, $v_4 = (1, -1)/\sqrt{2}$. Then $F_\mu = \{x \in \mathbb{R}^4 : x_1 + x_2 = x_3 + x_4\}$ and $d = 3$. The kernel at $v_0 = v_1$ is $k = (3, -1, 1, 1)$, and

$$\mathbb{E}_\mu[k] = \tfrac{1}{4}(3 - 1 + 1 + 1) = 1, \qquad K(v_0, v_0) = \|k\|^2 = \tfrac{1}{4}(9 + 1 + 1 + 1) = 3 = d,$$

$$\mathbb{E}_\mu[k\, vv^\top] = \tfrac{1}{4}\big(3v_1v_1^\top - v_2v_2^\top + v_3v_3^\top + v_4v_4^\top\big) = v_1v_1^\top \quad \text{exactly rank one.}$$

Normalising, $\mathbb{E}_\mu|k| = 3/2$, $c = 2/3$ and $r = (2, -\frac{2}{3}, \frac{2}{3}, \frac{2}{3})$ with $\mathbb{E}_\mu|r| = 1$; the entropy chain is $0.14384 \le \log \mathbb{E}_\mu[r^2] = \log \frac{4}{3} = 0.28768 \le \log 3 = 1.09861$. Note that r is genuinely signed: it is negative at v_2.

7 Verification record

Each round packages the artifact with the problem statement and a source card carrying every cited quotation verbatim, strips all provenance, renames it neutrally, and stages it outside the repository; five referees then read only that package, in mutually isolated fresh contexts, under a fixed referee prompt. Between rounds a handling editor rules every finding UPHELD / OVERRULED / PEDANTIC / NEEDS SOURCE, and a reviser is bounded strictly by the upheld list.

Round	Artifact	Clean	Where every substantive defect landed
1	`0048-RKHS-1`	1 / 5	scope commentary (Remark C)
2	`r2`	0 / 5	scope commentary: a false measure, $1 - o(1)$ vs 0.683
3	`r3`	4 / 5	scope commentary: a missing argument on the matrix domain
4	`r4`	2 / 5	scope commentary: the text added to repair round 3
5	`r5`	—	not run

Twenty passes across four rounds, over five angles — general, statement-drift and scope, measure-theoretic, line-by-line recomputation, and refutation-plus-source-fidelity — and three models, with the model rotated against the angle between rounds.

What the record shows. No pass in any round has faulted a load-bearing step. Lemmas 3–6, Theorem 1, Corollary 2 and Remark 7 are byte-identical across all five revisions, verified by hashing those regions at every boundary, and Corollary 8 is identical from `r3` onward. The twelve clean readings of that text are therefore readings of the same bytes. Four passes attempted explicit refutation — on both domains, with degenerate, atomic, discrete and continuous measures — and all four failed and said so. Several independently recomputed the kernel construction, the trace identity (2), the worked example and the constant $2/(e\delta)$, and found them correct.

What keeps failing. Every substantive defect in four rounds has been in the scope commentary of §5, never in §3. The failure mode has been stable: a repair to a limiting claim reaches slightly wider than the source card supports, and the next round catches it. Round 2's defect was a false number that a previous handling editor certified and instructed the reviser not to re-derive — which is the mechanism the multi-round structure exists to catch, working one stage later than it should have. Round 4's defect was the text added to repair round 3. That is a prose-calibration loop, not a mathematical one, and `r5` accordingly repairs by deletion alone, adding no replacement wording.

8 Provenance, and what would make this stronger

The passes were not blind in the strict sense. The harness specifies each pass as an isolated operating-system process with an empty working directory, so that blindness is a property of the process rather than a promise. This host has no model backend — no CLI binary and no API credentials — so the passes ran instead as in-session subagents. They get genuinely fresh contexts and never see the generator's reasoning, but they are one vendor and they run with the repository as their working directory. Mitigations applied: the package is staged outside the repository under a name unrelated to the artifact, all provenance and revision history is stripped, the referees are instructed not to read anything else, and every package is checked against a pattern list for leaked campaign machinery before dispatch. Each tally file carries this caveat in its header, and such rows must never be pooled with rows from a true `--bare` run.

Three things would materially raise confidence. First, re-running all five passes as isolated processes on a host with a backend, which is a matter of credentials rather than of work. Second, at least one pass from a different model family, which the harness recommends and which no round here has had: twenty same-vendor passes can share a blind spot that one cross-family pass would expose. Third, formalisation — the proof is short, elementary and almost entirely finite-dimensional linear algebra plus one Jensen inequality, which makes it an unusually good candidate for a proof assistant, and a machine-checked Lemma 3(b) and (2) would settle the two steps referees probed hardest.

Status of the record. Five artifacts, twenty referee reports, four editor rulings, a quotation-grounding result and the orchestrating script all sit in `c/0048/campaign/`, uncommitted. The orchestrator `scripts/verify-loop.sh` was written during this campaign because it was absent, and its first version returned the wrong termination code: it detected a blocking source defect by matching the prose "load-bearing", a phrase the referee prompt itself contains, so referees echoing or denying it tripped the test. It now tests the findings table structurally. That script has not itself been reviewed.